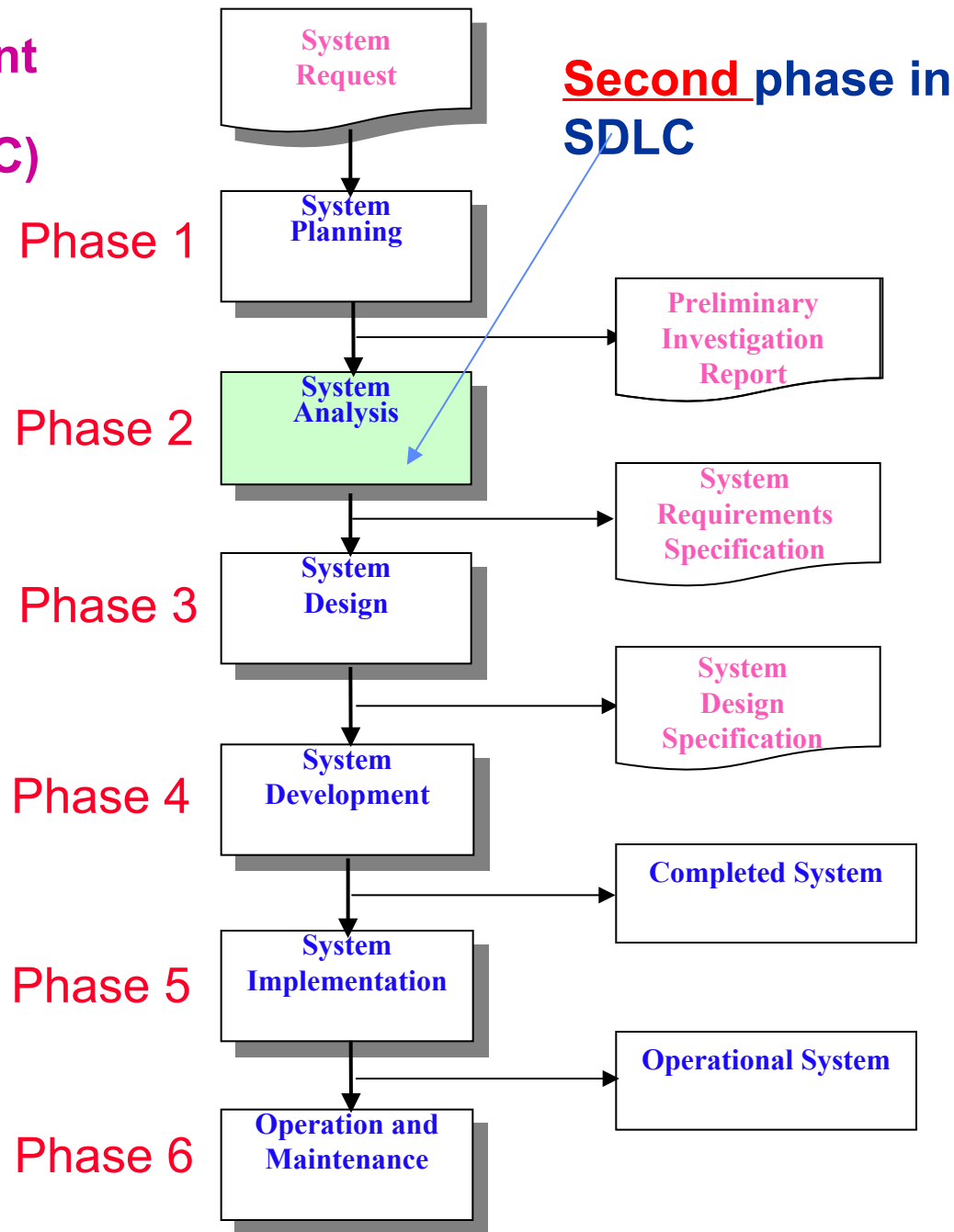


Chapter 5

System Analysis 2

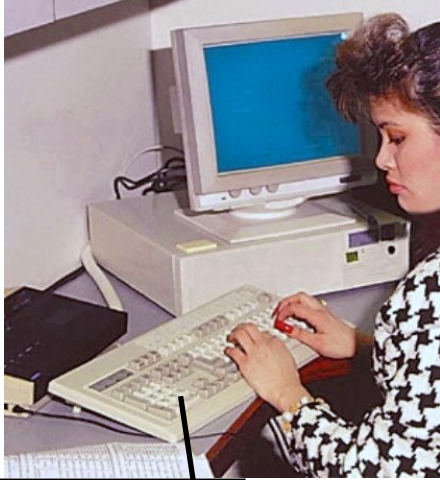
- Fact Recording 2

System Development Life Cycle (SDLC)



Entity Relationship Diagram (ERD)

Computerised system /information system



Manual system (without computer)

Employee form

Form 1: Employee Personal Information
Name of Department: _____

Employee Personal Information

First Name: _____ Middle Name: _____ Last Name: _____
Date of Birth: _____ Photo: 
Father/Mother/Husband Name: _____
Gender: ☐ male ☐ female Marital Status: _____
Identity Mark: _____

Mark the attached documents

☐ Medical Fitness ☐ Character Certificate

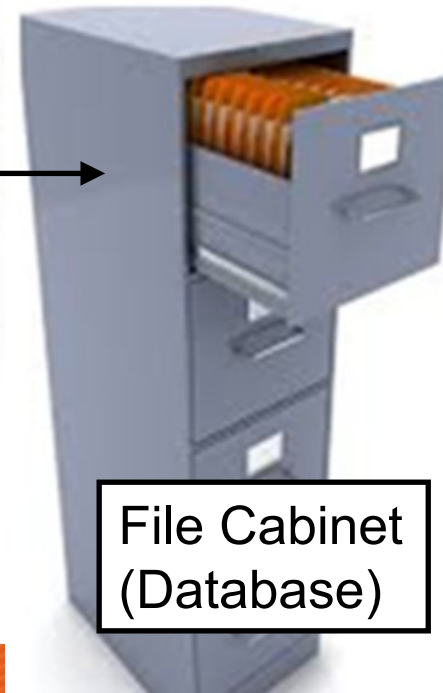
Height (in cm): _____
Caste: _____ Category: _____
Religion: _____ Blood Group: _____
Home State: _____ Home District: _____
Home Office Type: _____ Home Office Name: _____
LIC Home Town: _____ Nearest Railway St: _____
Awards (if any): _____

Employee office Details

Current Designation: _____ Current Office: _____
Current Code: _____



Employee file



File Cabinet
(Database)

Database

Employee file, customer file,
Order/Sale file ...

Employee Table / File

New Employee Collect Data via E-mail Add From Outlook E-mail List Reports						
ID	First Name	Last Name	E-mail Address	Business Phone	Company	Job Title
1	Nancy	Freehafer	nancy@northwindtraders.com	(123)555-0100	Northwind Trade	Sales Representative
2	Andrew	Cencini	andrew@northwindtraders.com	(123)555-0100	Northwind Trade	Vice President, Sales
3	Jan	Kotas	jan@northwindtraders.com	(123)555-0100	Northwind Trade	Sales Representative
4	Mariya	Sergienko	mariya@northwindtraders.com	(123)555-0100	Northwind Trade	Sales Representative
5	Steven	Thorpe	steven@northwindtraders.com	(123)555-0100	Northwind Trade	Sales Manager
6	Michael	Neipper	michael@northwindtraders.com	(123)555-0100	Northwind Trade	Sales Representative
7	Robert	Zare	robert@northwindtraders.com	(123)555-0100	Northwind Trade	Sales Representative
8	Laura	Giussani	laura@northwindtraders.com	(123)555-0100	Northwind Trade	Sales Coordinator
9	Anne	Hellung-Larsen	anne@northwindtraders.com	(123)555-0100	Northwind Trade	Sales Representative

Purchase request (data flow --> data)

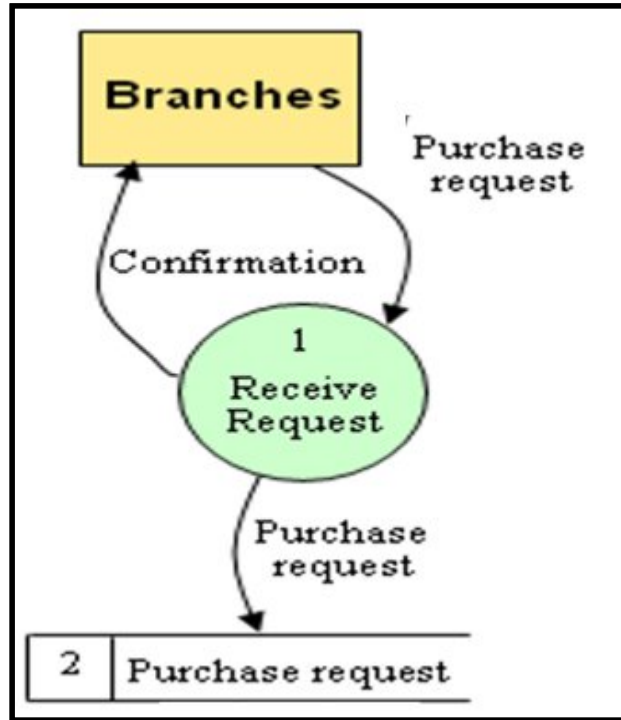
Last Monday -Perak - Ali request to buy 10 boxes - A4 papers (500 sheets per package), 1 box -4 pack	accept
--	--------

Last Thursday - Johor - David request to buy 20 boxes - A4 papers (500 sheets per package), 1 box -4 pack	accept
---	--------

Purchase Request (data store --> file)

PR01 -Last Monday -Perak Ali request to buy 10 boxes - A4 papers (500 sheets per package), 1 box -4 pack

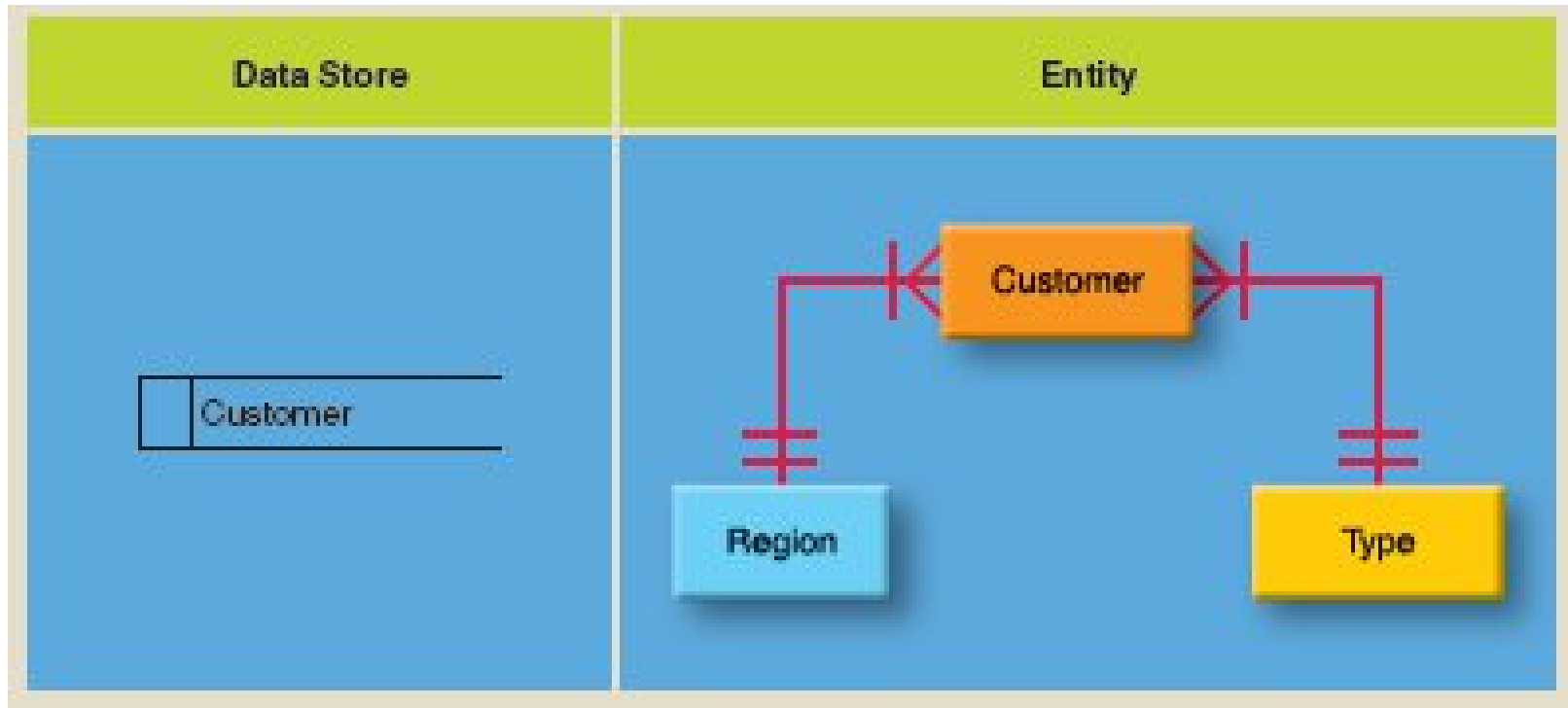
PR02 -Last Thursday - Johor - David request to buy 20 boxes - A4 papers (500 sheets per package), 1 box -4 pack



Data Flow Diagram
(DFD)

Relationship Between DFD and ERD

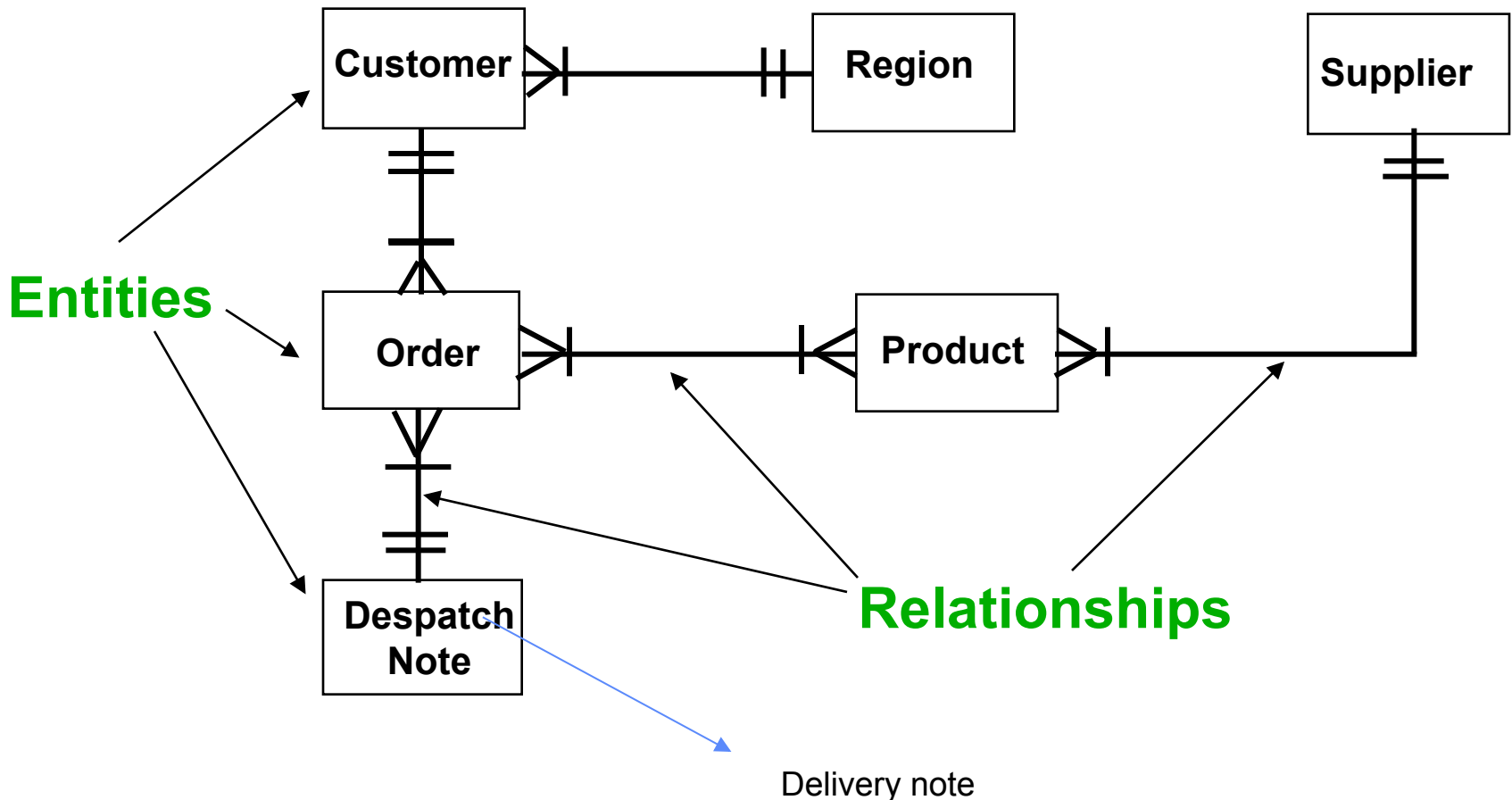
(cont...)



The **data stores** in DFD is related to **entities** in ERD. The number of data stores can be used as a starting point for drawing the ERD. But take note that this relationship may not be 1 : 1.

Components of ERD

- There are only **2** components :



ERD Notations

	Crow's Foot	Chen
Entity		
Relationship line		
Relationship	None	
Option Symbol		
One (1) symbol		1
One and only one symbol		1
Many (M) symbol		M (or N)

Note : This University adopts the *Crow's Foot* notation for all its *tests*, *assignments* and *examination*, for this course.

Entity (= file, table, data stored)

- **Meaning.** A class of persons, places, branches, departments, objects, or transactions. They store data which are entered or captured. Described using a noun.
- **Examples :** customers, products, suppliers, Purchase Order (PO), payment, Sale Order (SO), quotations.
- **Notation.** It is represented by a rectangular box.



Examples of Entities Types

- **Persons** : contractor, customer, employee, student, supplier, publisher, applicant.
- **Objects** : book, machine, stock (part, product, item), course, programme.
- **Events/Transactions** : application, reservation, cancellation, class, flight, invoice, registration, payment, renewal, manpower requisition, reservation, Sale Order, Purchase Order, Request For Quotation (RFQ) , Quotation, Request Purchase
- **Places** : sales region, building, room, branch office, campus, department, faculty

- *Central Region*
 - Kuala Lumpur (federal territory)
 - Putrajaya (federal territory)
 - Selangor
- *Northern Region*
 - Kedah
 - Penang
 - Perak
 - Perlis
- *Southern Region*
 - Johor
 - Melaka
 - Negeri Sembilan

Exercise 2

- List 4 appropriate **entities** for each of the following systems :
 - Inventory System
 - Recruitment System
 - Purchasing System

Answers 2

- **Inventory System**
 - Supplier, Item, Goods Delivered, Goods Received etc
- **Recruitment System**
 - Department, Manpower Requisition, Application, Applicant, Interview, Job etc
- **Purchasing System**
 - Supplier, Purchaser, Item, Purchase Order, Quotation, Request For Quotation etc

Note : the answers above are NOT exhaustive, just samples only

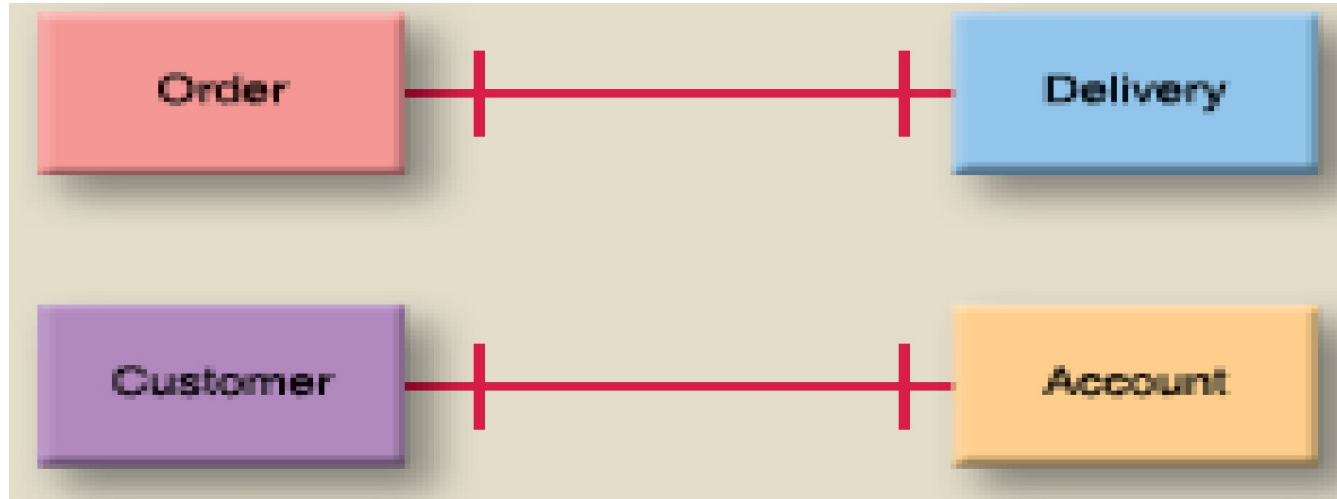
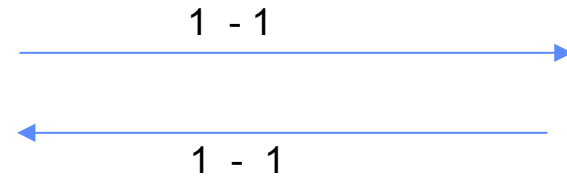
Relationship

- **Meaning.** The lines between the entities (boxes) represent relationships.
- **Example.** Between **SUPPLIER** and **PURCHASE ORDER**
- **Notation.** The relationship is indicated by **lines** connecting the entities



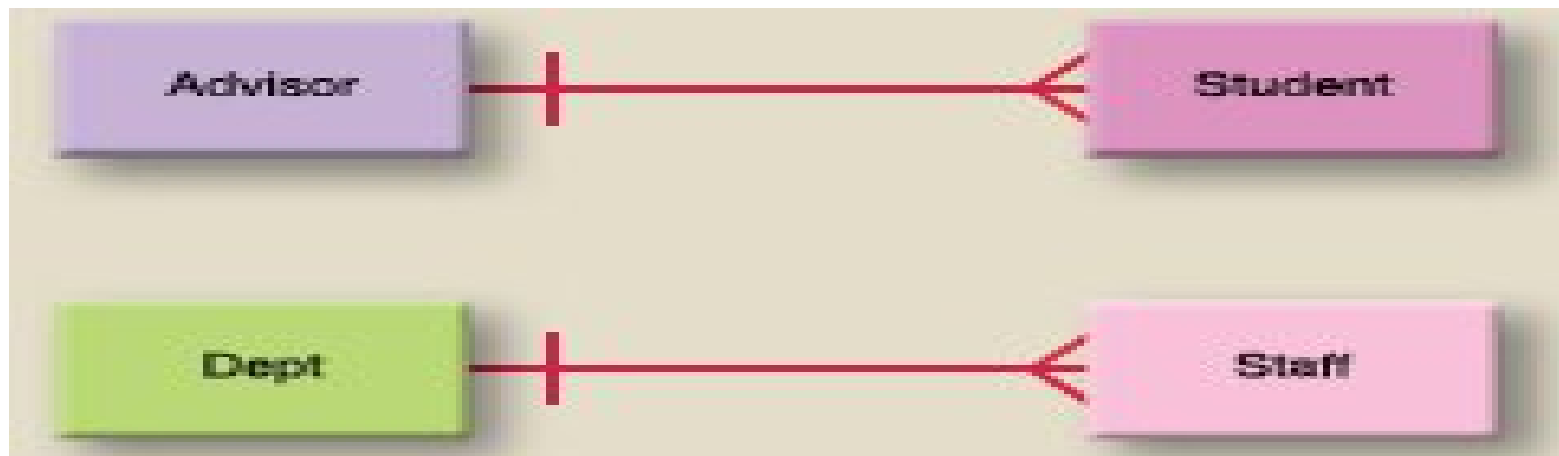
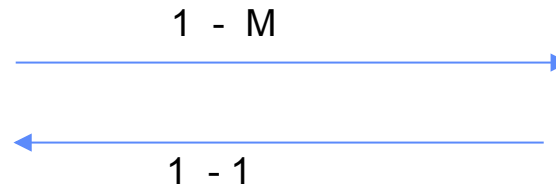
3 Types of Relationship

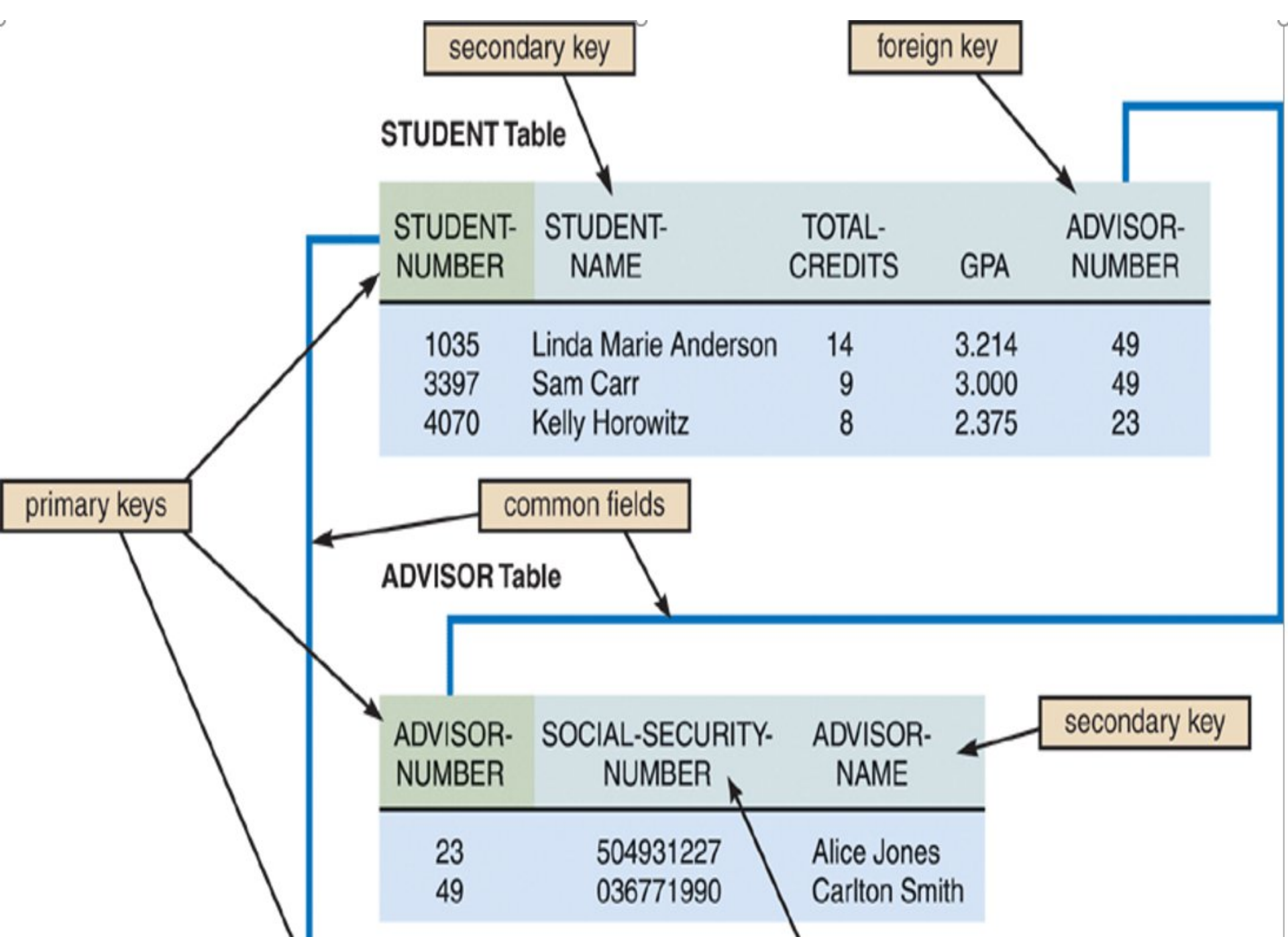
- One-to-One (1 : 1)



User account

- One-to-Many (1 : N)





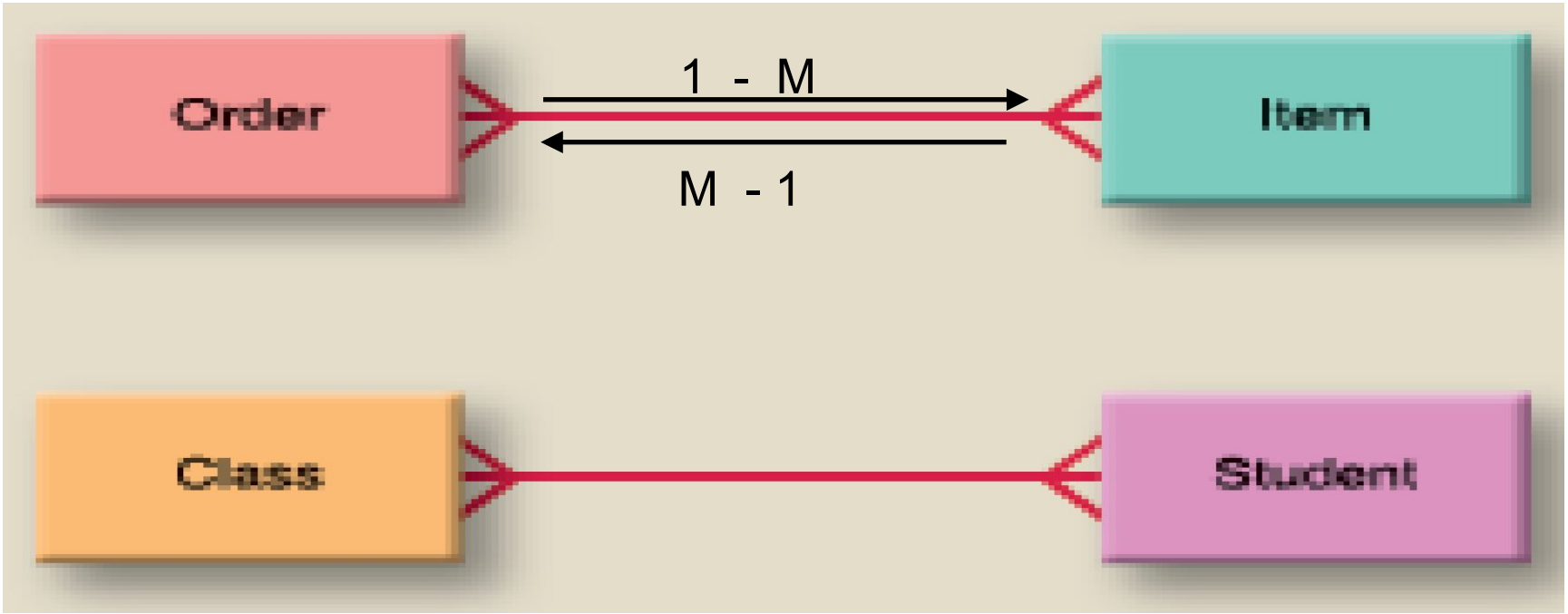
Types of Relationship

- Many-to-Many (M : N)

Item				
I001	item A	RM 10	100	units
I002	Item B	RM 5	500	units
I003	Item C	RM 5	300	units

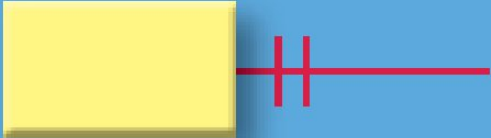
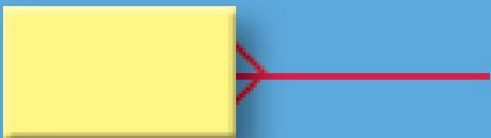
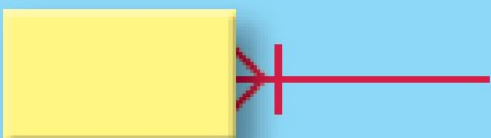
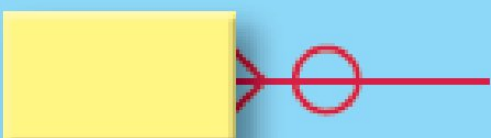
order/sale/sale order
Date : 07/06
Customer :C01, Ali
Order No : **O001**
I001 item A RM10 10 units
I002 Item B RM5 20 units

order/sale/sale order
Date : 08/06
Customer:C01, Ali
Order No : **O002**
I001 item A RM10 20 units
I003 Item C RM5 30 units



Cardinality – no. of occurrences

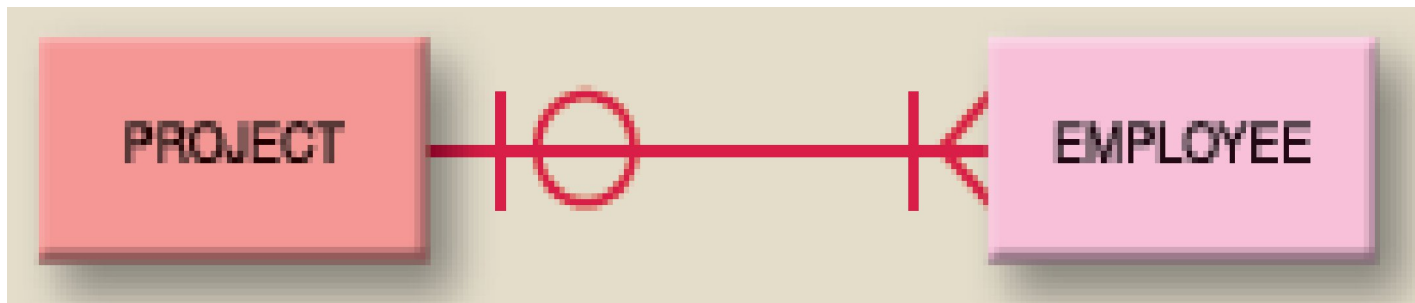
Crow's Foot Notation

Symbols	Meaning
	One and only one
	Zero or one
	More than one
	one or more than one
	Zero or one or more than one (ie. More than zero)

Note : This University adopts the Crow's Foot notation for all its tests, assignments and examination, for this course.

Crow's Foot Notation

- Describes the **type** of relationship between two entities
- Cardinality must be defined in **both directions** for every relationship.

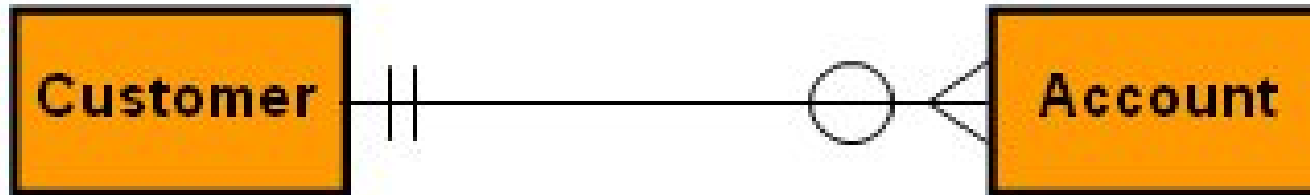


1st direction —→ One **Project** has *One or Many* **Employee**

2nd direction —→ One **Employee** takes part in *Zero or One* **Project**

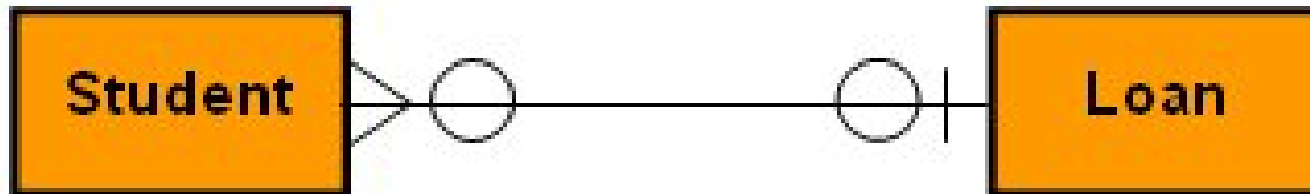
Crow's Foot Notation Examples

Credit card account



One ~~and only one~~ CUSTOMER can open zero or one or many ACCOUNT.

One ACCOUNT is opened by only one CUSTOMER



Student
Loan

One STUDENT can apply for zero or one LOAN.

One LOAN is applied by zero or one, or many STUDENT

Example 1 – Sales System

Customer Buys Products /Goods/Items

KL Central Region
Selangor Central Region

order/sale/sale order
Date : 07/06
Customer :C01, Ali
Order No : **O001**
I001 item A RM10 10 units
I002 item B RM5 20 units

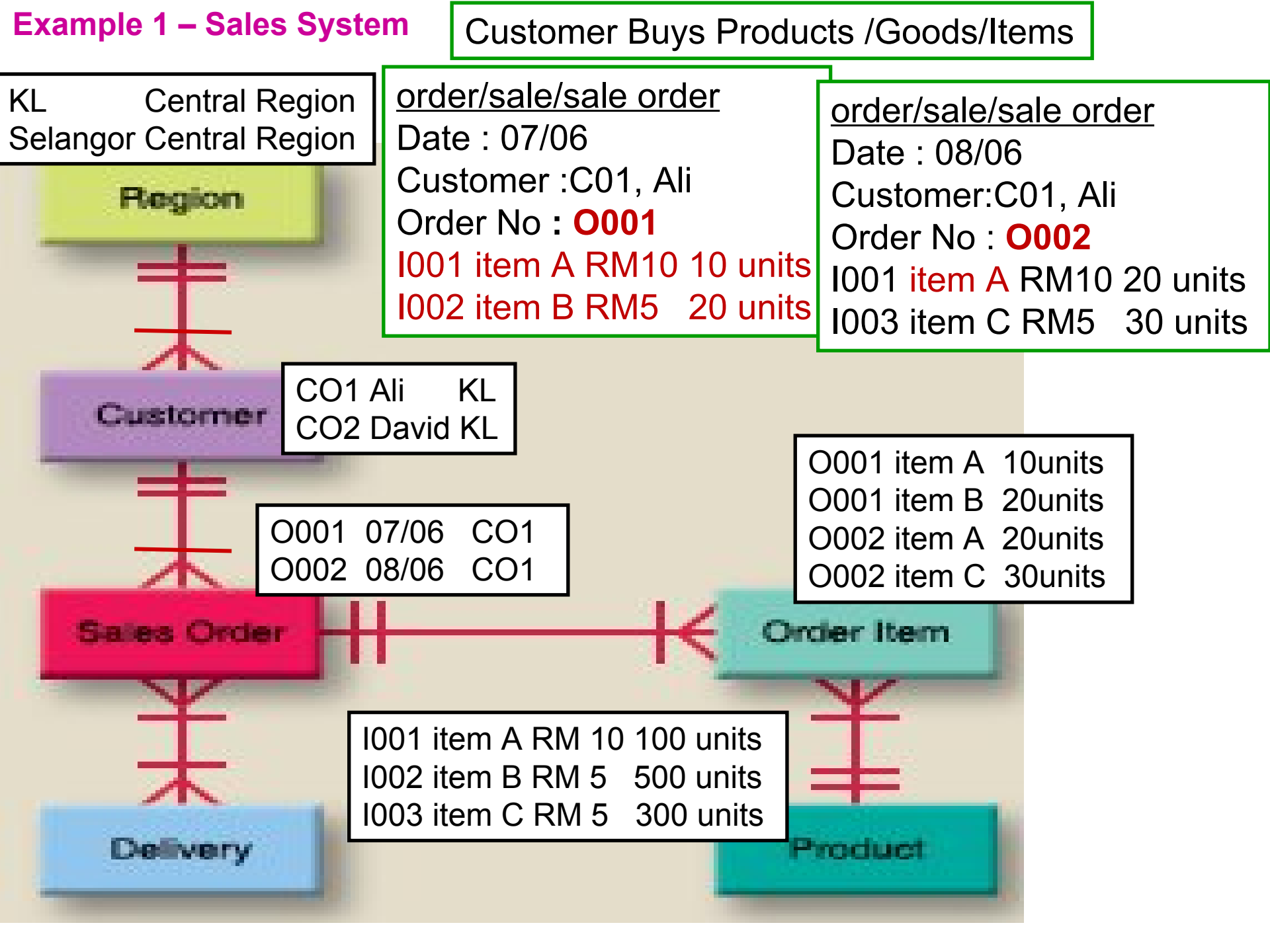
order/sale/sale order
Date : 08/06
Customer:C01, Ali
Order No : **O002**
I001 item A RM10 20 units
I003 item C RM5 30 units

CO1 Ali KL
CO2 David KL

O001 07/06 CO1
O002 08/06 CO1

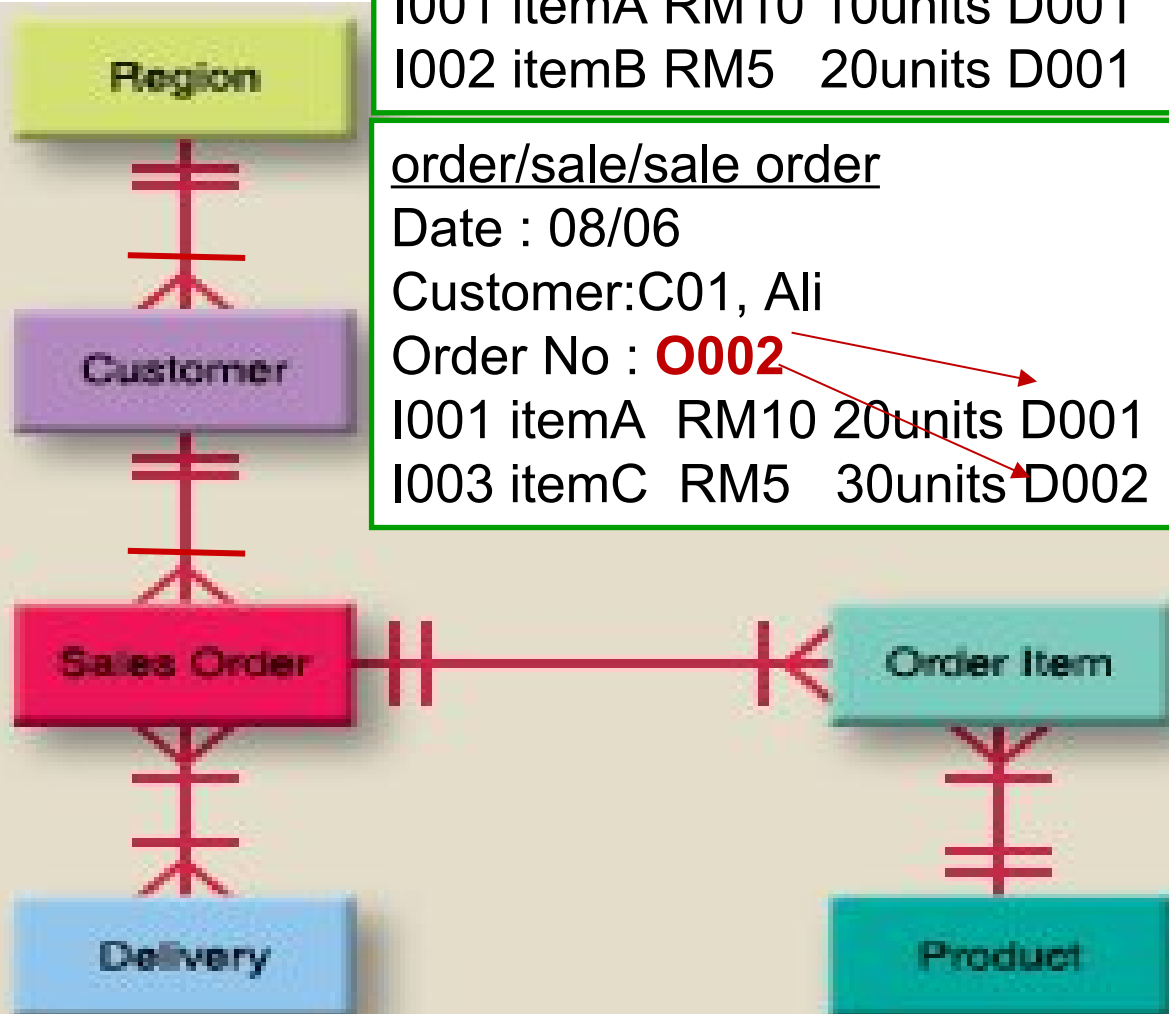
O001 item A 10units
O001 item B 20units
O002 item A 20units
O002 item C 30units

I001 item A RM 10 100 units
I002 item B RM 5 500 units
I003 item C RM 5 300 units



Example 1 – Sales System

Customer Buys Products /Goods/Items



order/sale/sale order

Date : 07/06

Customer :C01, Ali

Order No : **0001**

I001 itemA RM10 10units D001

I002 itemB RM5 20units D001

order/sale/sale order

Date : 08/06

Customer:C01, Ali

Order No : **0002**

I001 itemA RM10 20units D001

I003 itemC RM5 30units D002

Delivery

Delivery No : **D001**

Date:09/06

Staff : David

Customer :C01, Ali

Order No : **0001**

I001 itemA RM10 10units

I002 itemB RM5 20units

Order No : **0002**

I001 itemA RM10 20units

Delivery

Delivery No : **D002**

Date:11/06

Staff : David

Customer :C01, Ali

Order No: **0002**

I003 itemC RM5 30units

Example 2 - Inventory System

Customer Buys Products /Goods/Items

order/sale/sale order

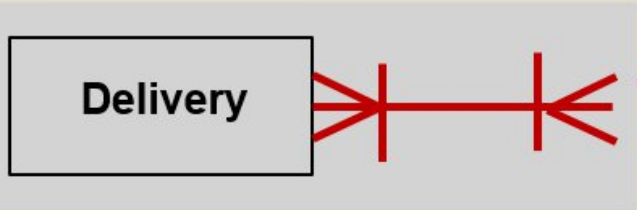
Date : 07/06

Customer :C01, Ali

Order No : **O001**

I001 itemA RM10 10units

I002 itemB RM5 20units



Return Inwards

ReturnInwardsNo: RI001

Date : 14/06

Customer :C01, Ali

Order No : **O001**

I001 itemA 5units

ReturnInwardsNo: RI002

Date : 15/06

Customer :C01, Ali

Order No : **O001**

I002 Item B 1 units

ReturnInwardsNo: RI003

Date : 18/06

Customer :C01, Ali

Order No : **O001**

I002 Item B 2 units

Purchase request (data flow --> data)

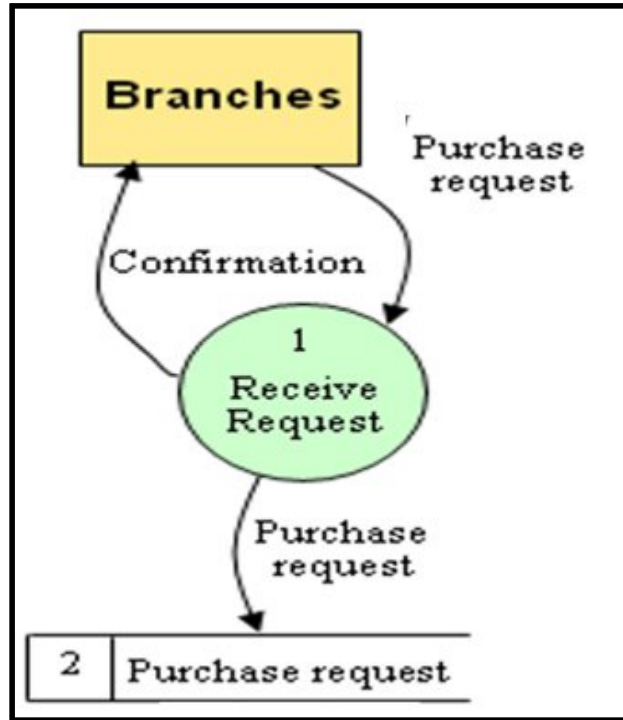
Last Monday -Perak - Ali request to buy 10 boxes - A4 papers (500 sheets per package), 1 box -4 pack	accept
--	--------

Last Thurday - Johor - David request to buy 20 boxes - A4 papers (500 sheets per package), 1 box -4 pack	accept
--	--------

Purchase Request (data store --> file)

PR01 -Last Monday -Perak Ali request to buy 10 boxes - A4 papers (500 sheets per package), 1 box -4 pack

PR02 -Last Thurday - Johor - David request to buy 20 boxes - A4 papers (500 sheets per package), 1 box -4 pack



Data Flow Diagram
(DFD)

Purchase request (data flow --> data)

Last Monday - Perak - Ali request to buy 10 boxes - A4 papers (500 sheets per package), 1 box - 4 pack	accept
--	--------

Last Thursday - Johor - David request to buy 20 boxes - A4 papers (500 sheets per package), 1 box - 4 pack	accept
--	--------

Purchase Request (data store --> file)

PR01 - Last Monday - Perak Ali request to buy 10 boxes - A4 papers (500 sheets per package), 1 box - 4 pack

PR02 - Last Thursday - Johor - David request to buy 20 boxes - A4 papers (500 sheets per package), 1 box - 4 pack

PO: PO001 **Supplier A** Date :5/08/2024

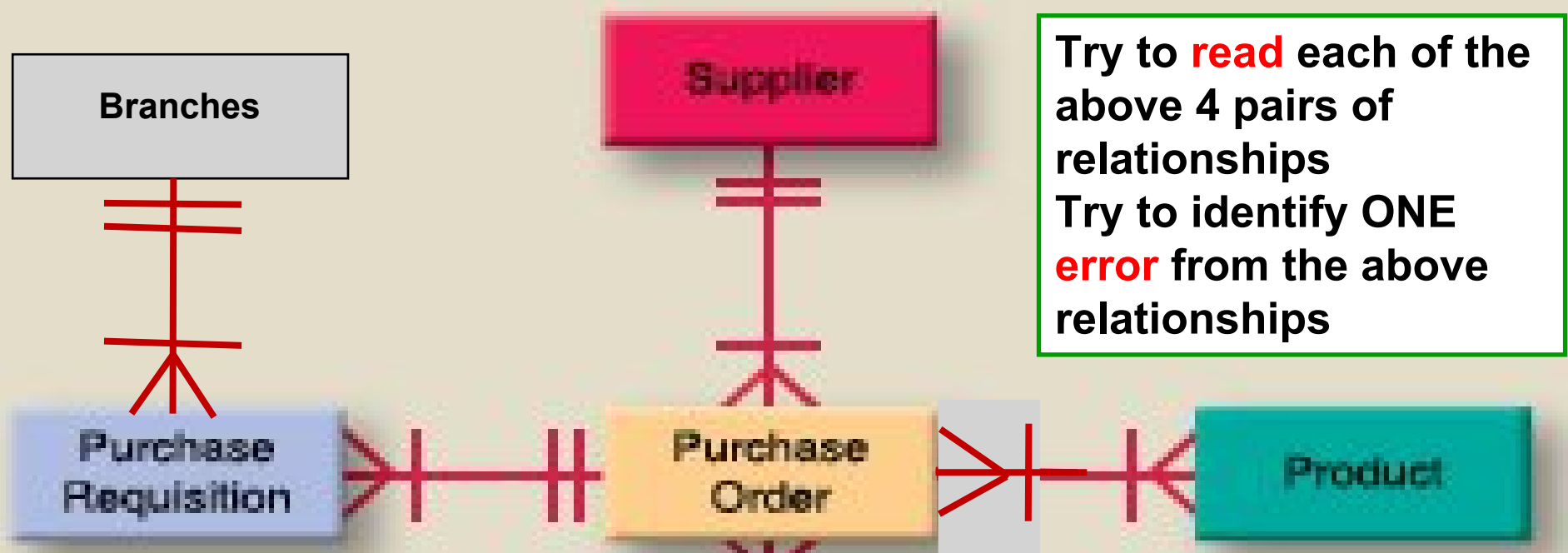
Buy ~~30~~ boxes A4 papers [1 box - 4 pack] RM2000

Buy ball pen (blue) 20 boxes [1 box - 50pcs] RM200

PO: PO002 **Supplier A** Date :08/08/2024

Buy ball pen (blue) 20 boxes [1 box - 50pcs] RM200

Example 3 – Purchase Order Processing



Try to **read** each of the above 4 pairs of relationships
Try to identify **ONE error** from the above relationships

Purchase Request (data store --> file)

- PR01 -Last Monday -Perak Ali request to buy 10 boxes - A4 papers (500 sheets per package), 1 box -4 pack
- PR02 -Last Thurday - Johor - David request to buy 20 boxes - A4 papers (500 sheets per package), 1 box -4 pack

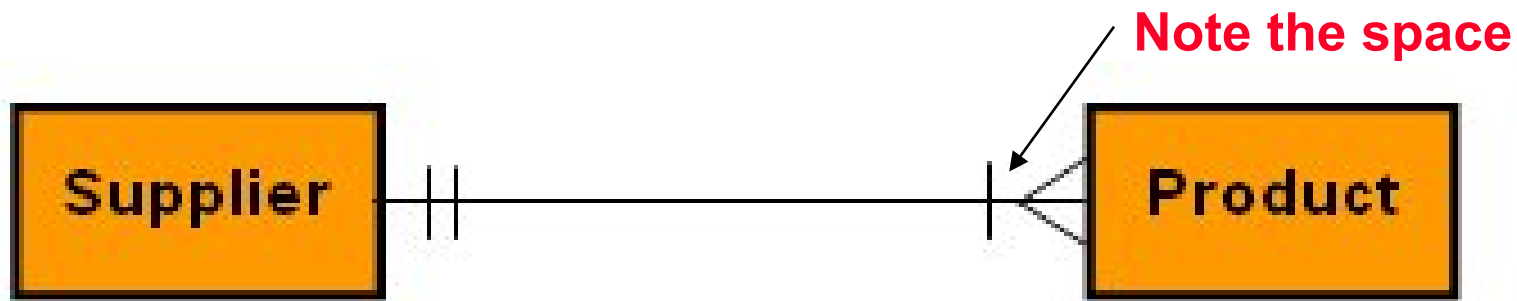
PO: PO001 **Supplier A** Date :5/08/2024
Buy **30** boxes A4 papers [1box - 4 pack] RM2000
Buy **ball pen (blue)** 20 boxes [1 box -50pcs] RM200

PO: PO002 **Supplier A** Date :08/08/2024
Buy **ball pen (blue)** 20 boxes [1 box -50pcs] RM200

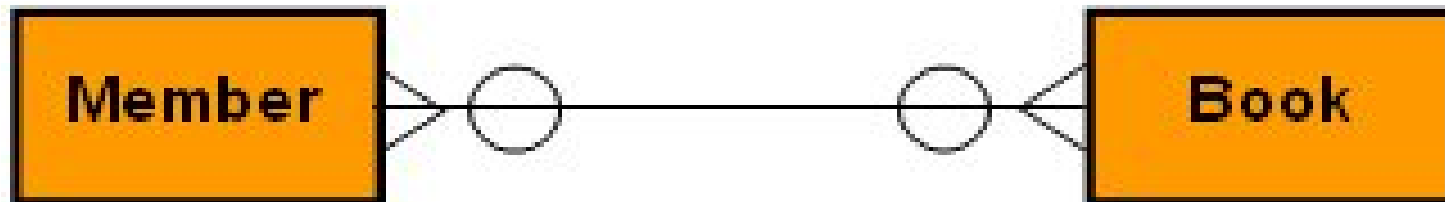
Resolving Many-to-Many Relationship

(**Note** : any many-to-many relationships must be resolved)

Crow's Foot Notation Examples



One SUPPLIER can supply one or many PRODUCT.
One PRODUCT can be supplied by only one



One MEMBER can borrow zero or one or many BOOK
One BOOK can be borrowed by zero or one or many
MEMBER

Example 1 – Associative Entity

- By introducing an 'associative' entity to form a new 2 pairs of one-to-many relationships.

Customer Buys Products /Goods/Items

order/sale/sale order complete

Date : 07/06

Customer :C01, Ali

Order No : **O001**

Total amount : RM180

Grand total amount : RM 160

I001itemA RM10X10units =RM100 RM80

I002itemB RM5X20 units =RM100 RM100

(item final price -after discount price, item A got 20% discount)

order/sale/sale order

Date : 08/06

Customer:C01, Ali

Order No : **O002**

I001item A RM10 x 20 units xxx xxx

I003item C RM5 x 30 units xxx xx

(final order price -after discount price for each order)
>RM150 , -RM20 (Discount)

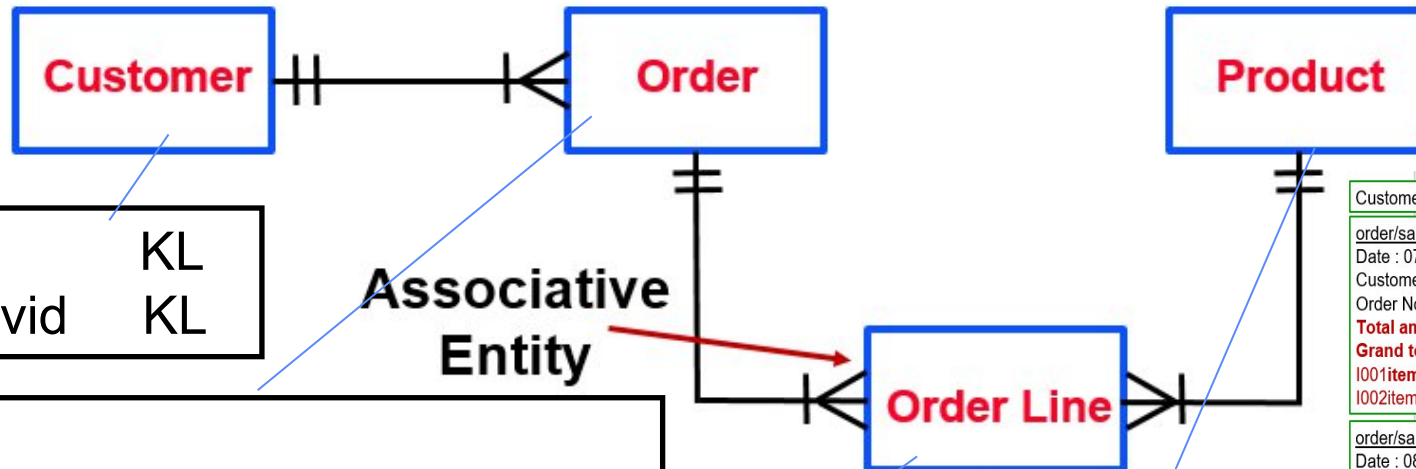
Example 1 – Associative Entity

- By introducing an 'associative' entity to form a new 2 pairs of one-to-many relationships.

Before



After



CO1 Ali KL
CO2 David KL

Associative
Entity

Order

O001 07/06 CO1 RM180 RM160
O002 08/06 CO1 XXX XXX:

Customer Buys Products /Goods/Items

order/sale/sale order **complete**
Date : 07/06
Customer : CO1, Ali
Order No : **O001**
Total amount : RM180
Grand total amount : RM 160
I001itemA RM10X10units =RM100 RM80
I002itemB RM5X20 units =RM100 RM100

order/sale/sale order
Date : 08/06
Customer:CO1, Ali
Order No : **O002**
I001item A RM10 x 20 units xxx xxx
I003item C RM5 x 30 units xxx xx

O001 I001 10units RM10 RM100 RM80
O001 I002 20units RM20 RM100 RM100
O002 I001 20units RM10 XXX XXX
O002 I003 30units RM5 XXX XXX

I001 itemA RM10 100 units
I002 itemB RM5 500 units
I003 itemC RM5 300 units

Listing Data Attributes (Fields)

Customer Buys Products /Goods/

order/sale/sale order

Date : 07/06

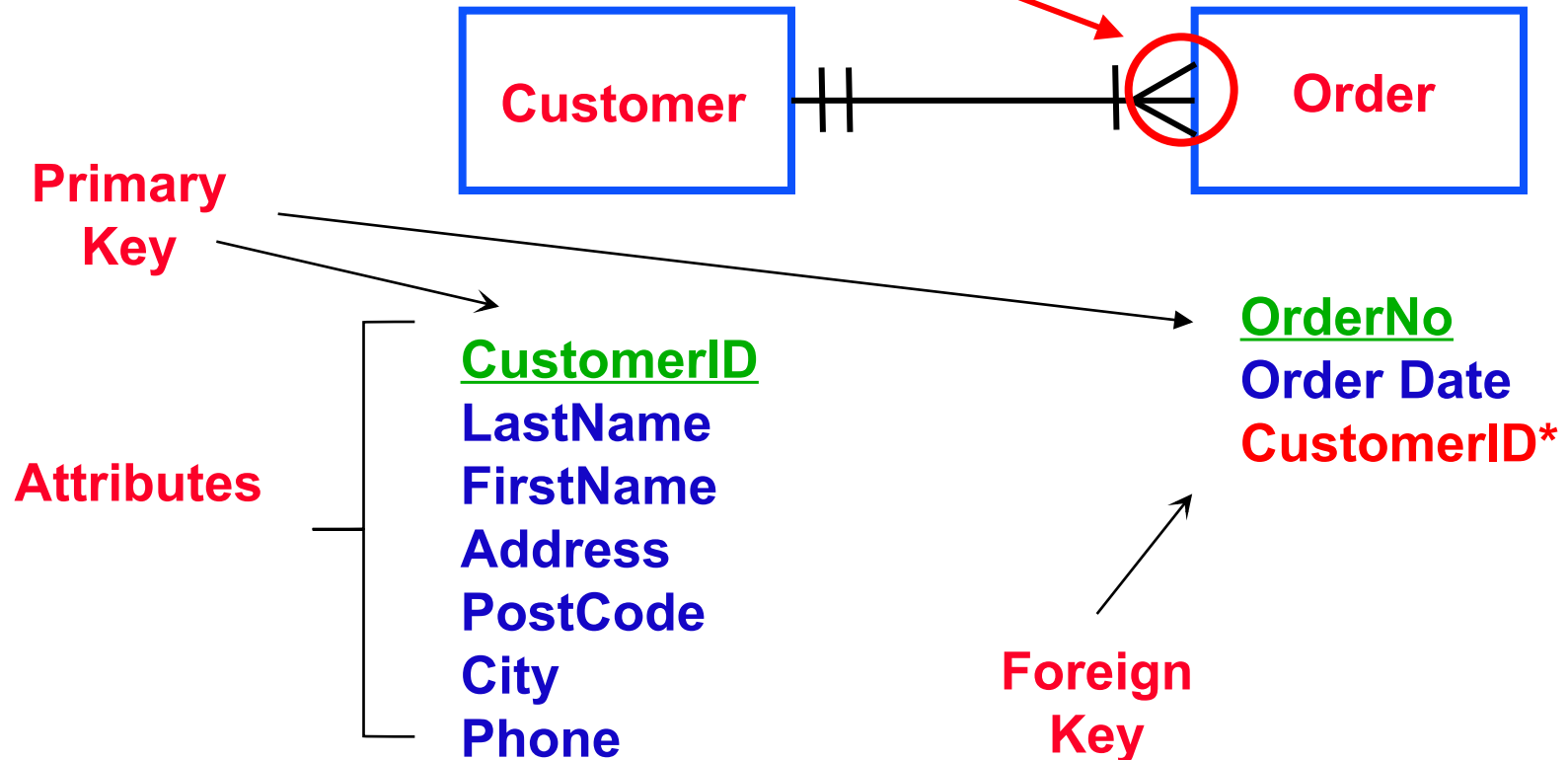
Customer :C01, Ali

Order No : **O001**

I001 item A RM10 10 units

I002 Item B RM5 20 units

Shows that this Entity
has a **Foreign Key**



Sample Data for Each Attribute

Table Name

Attribute Names

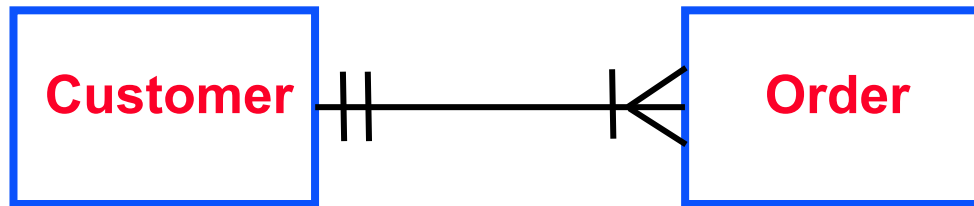
STUDENT

<u>STUDENT- NUMBER</u>	STUDENT- NAME	TOTAL- CREDITS	GPA	ADVISOR- NUMBER	ADVISOR- NAME
1035	Linda	47	3.647	49	Smith
3397	Sam	29	3.000	49	Smith
4070	Kelly	14	2.214	23	Jones

Data
records

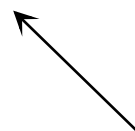
Listing Data Attributes in DBDL format

- DBDL = Database Design Language

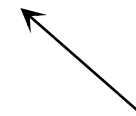


CUSTOMER (CustomerID, LastName, FirstName, Address, PostCode, City, Phone)

ORDER (OrderNo, Order Date, CustomerID^{*})



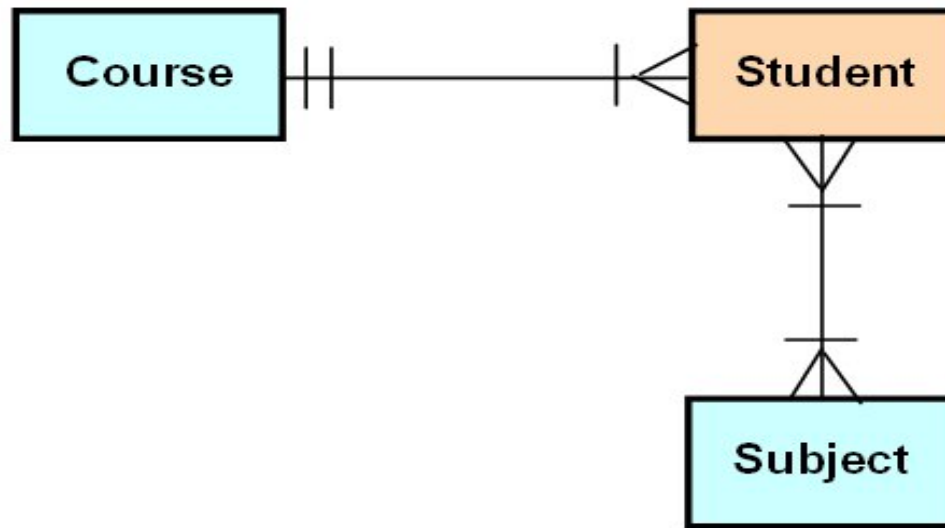
Primary Key



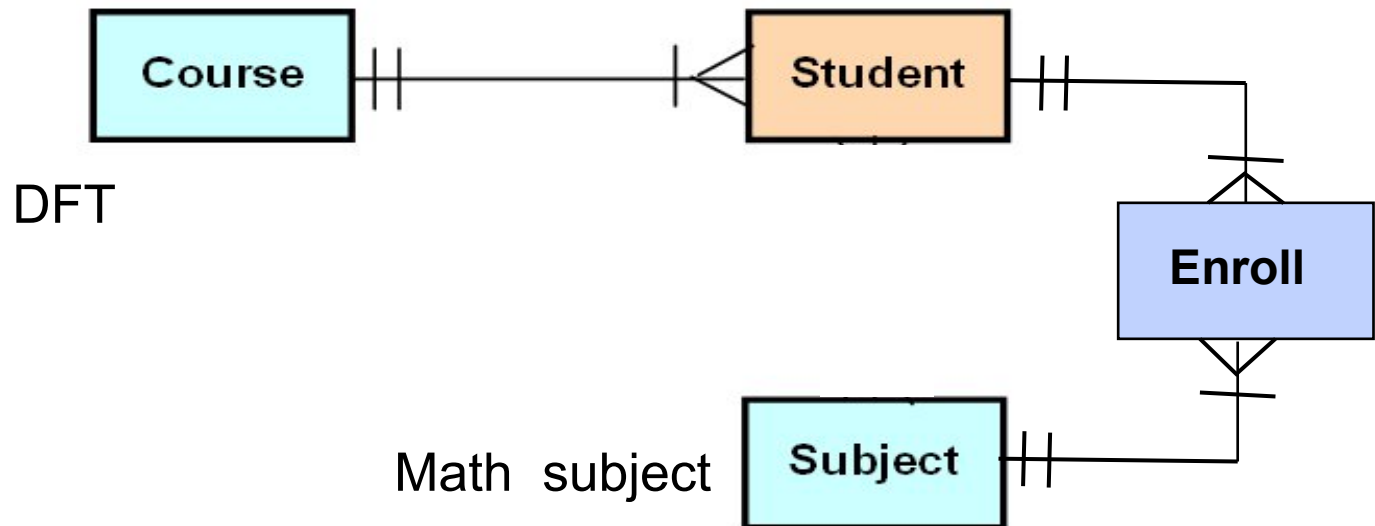
Foreign Key

Example 2

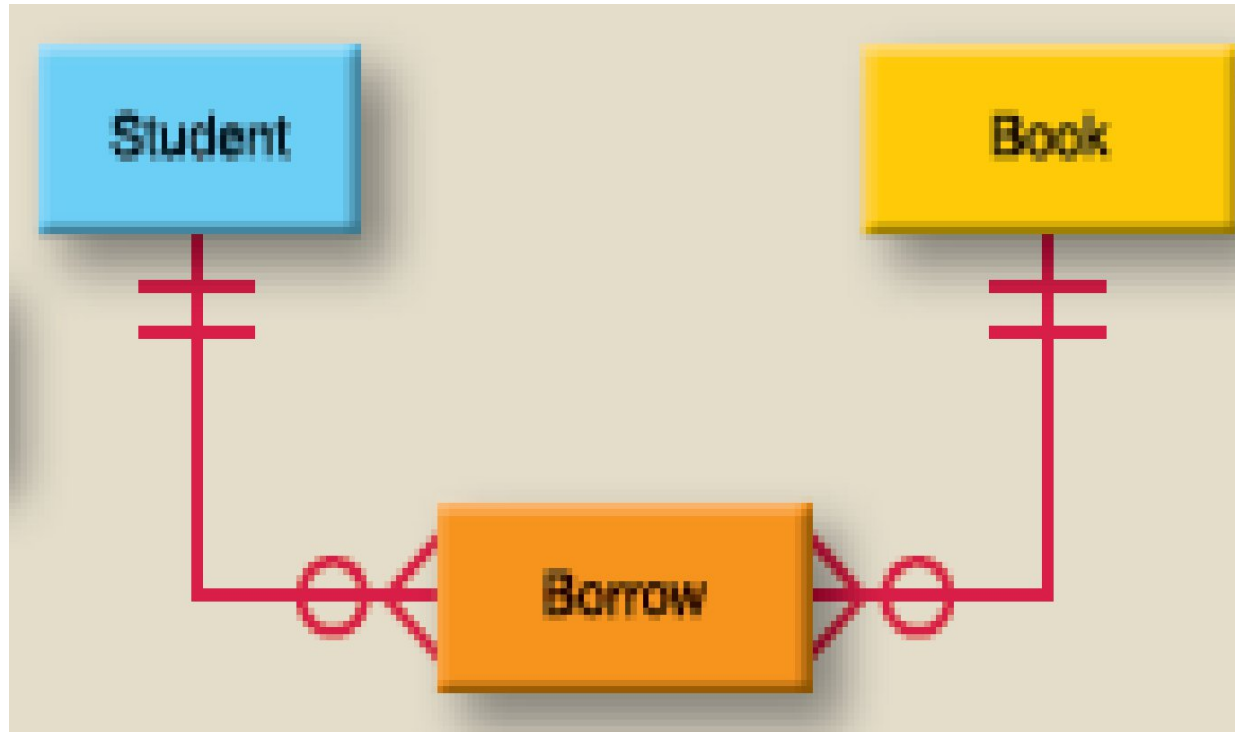
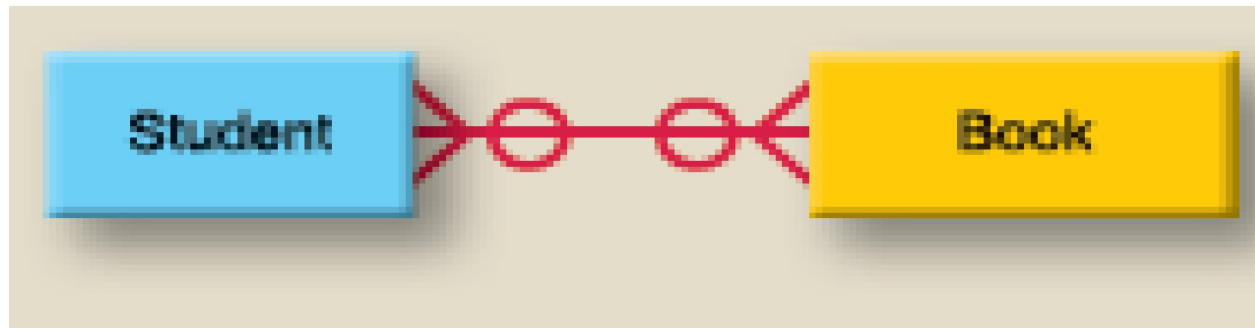
Before



After



Example 3



Purposes of ERD

- **Business Rules** – capture and describe the business rules and relationships between the entities.
- **Modeling System** – describe the business system in diagrams.
- **Users** – communicating system to the users for their confirmation (sign-off) of requirements.
- **File Design** – form the basis of subsequent data and file design. The entities become physical files or database tables.

- **File Design** – form the basis of subsequent data and file design. The entities become physical files or database tables.

employee

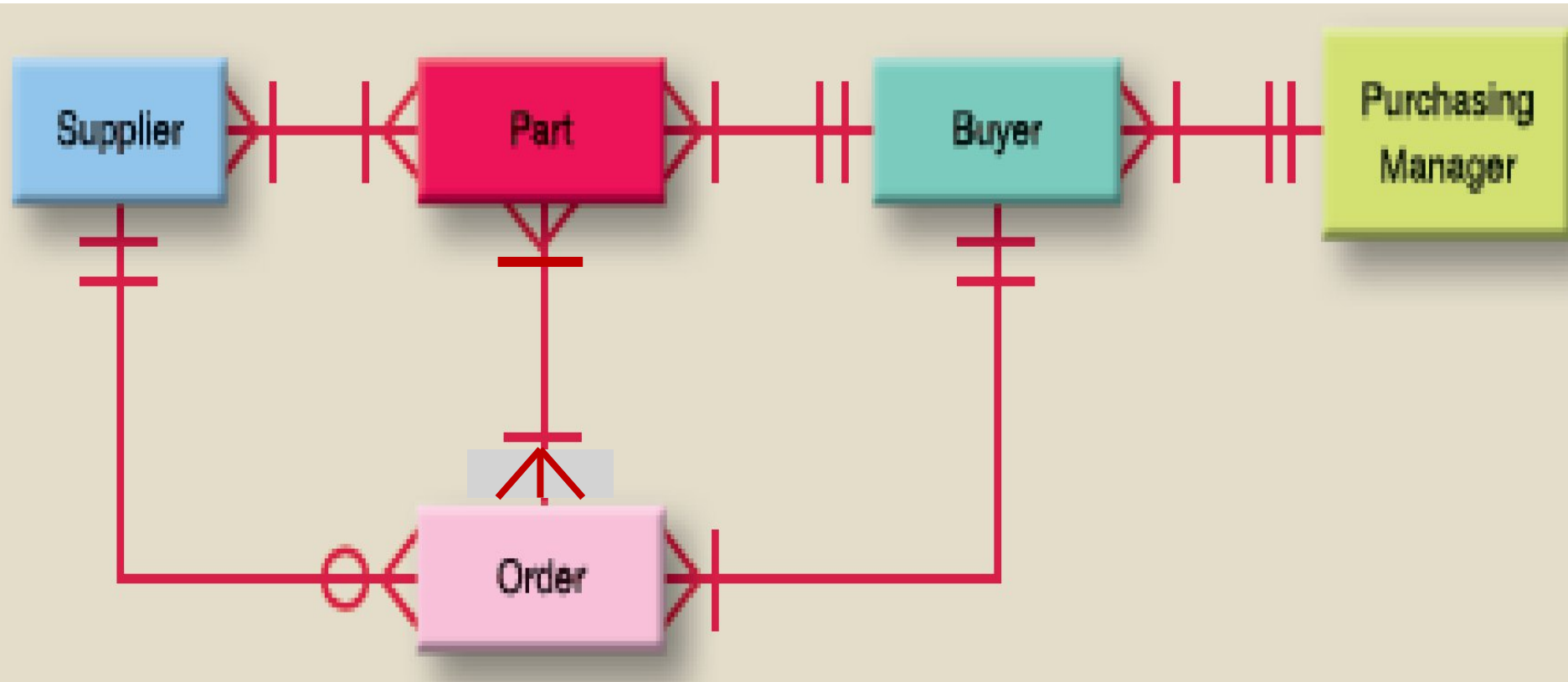
Employee Table / File						
New Employee Collect Data via E-mail Add From Outlook E-mail List Reports						
ID	First Name	Last Name	E-mail Address	Business Phone	Company	Job Title
1	Nancy	Freehafer	nancy@northwindtraders.com	(123)555-0100	Northwind Trade	Sales Representative
2	Andrew	Cencini	andrew@northwindtraders.com	(123)555-0100	Northwind Trade	Vice President, Sales
3	Jan	Kotas	jan@northwindtraders.com	(123)555-0100	Northwind Trade	Sales Representative
4	Mariya	Sergienko	mariya@northwindtraders.com	(123)555-0100	Northwind Trade	Sales Representative
5	Steven	Thorpe	steven@northwindtraders.com	(123)555-0100	Northwind Trade	Sales Manager
6	Michael	Neipper	michael@northwindtraders.com	(123)555-0100	Northwind Trade	Sales Representative
7	Robert	Zare	robert@northwindtraders.com	(123)555-0100	Northwind Trade	Sales Representative
8	Laura	Giussani	laura@northwindtraders.com	(123)555-0100	Northwind Trade	Sales Coordinator
9	Anne	Hellung-Larsen	anne@northwindtraders.com	(123)555-0100	Northwind Trade	Sales Representative
(New)						

Exercise 1 - Purchasing System

- A company has 10 buyers (purchasing staff), each one reporting to one of two purchasing managers.
- Each buyer has sole responsibility for the ordering of 500 of the 5000 parts/products/items used by the company.
- The same part can be supplied by several suppliers.
- Each purchase order can contain several parts.

Note : The above has been intentionally described in **ONE** direction only. Students should make logical assumption on the other direction.

Suggested Answer 1



invoice

LOGO

FROM

East Asia Trading
Pasar Pudu Baru 10
Kuala Lumpur 53000

INVOICE

MY-001

INVOICE DATE

29/01/2019

P.O.#

2330/2019

DUE DATE

24/05/2019

BILL TO

Mayang Bujang
2 Pasar Moden 55
Kuala Lumpur 55100

SHIP TO

Mayang Bujang
1721 Jln Sp 232/7
Kuala Lumpur 55103

QTY	DESCRIPTION	UNIT PRICE	AMOUNT
1	Wooden elephant figurine	600.00	600.00
2	Large cloth rice bag	45.00	90.00
3	Bamboo ladder	20.00	60.00

Invoice line / order product

Subtotal 750.00
SST 6.0% 45.00

TOTAL

RM795.00



TERMS & CONDITIONS

Payment is due within 15 days

Public Bank Berhad
Account Number: 12345678
Routing Number: 0987654321098

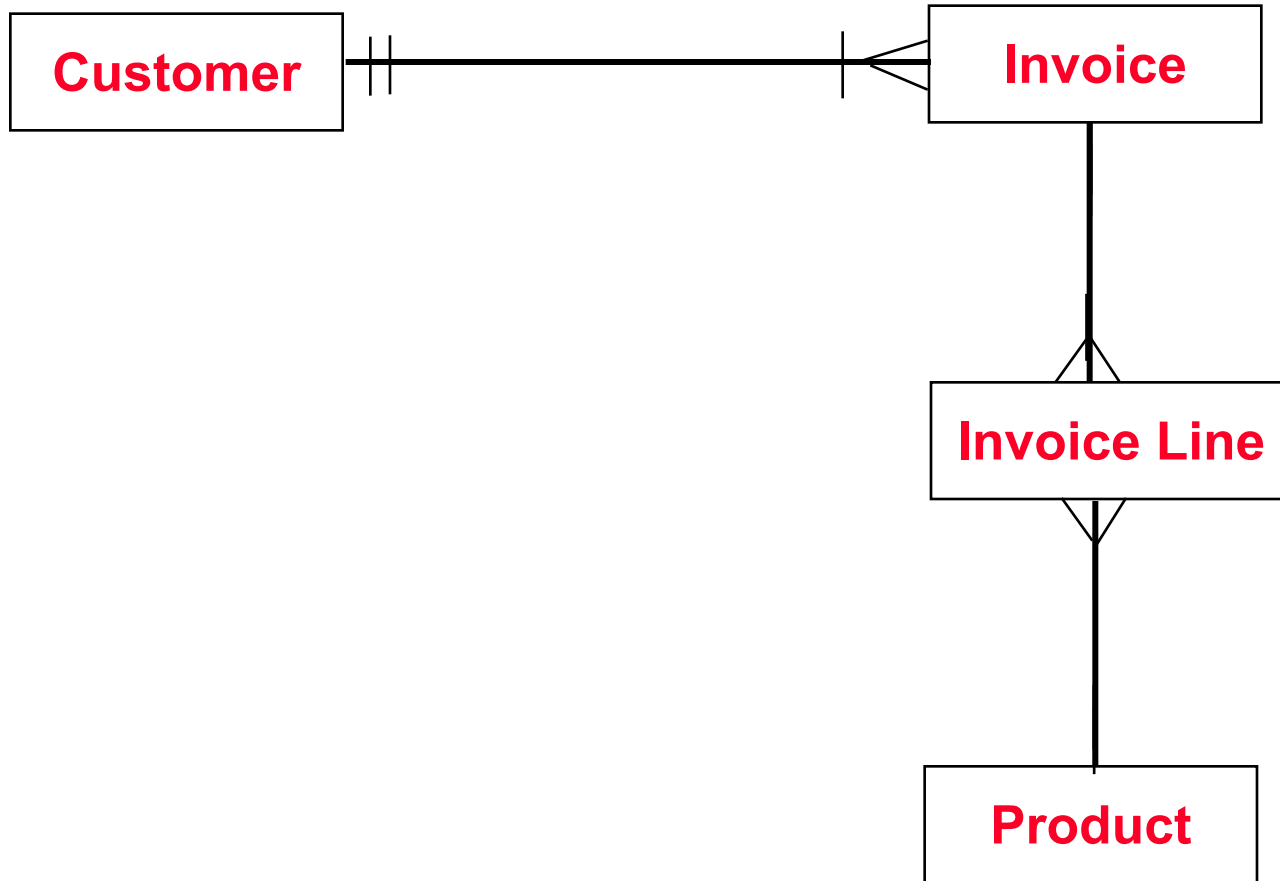
<https://invoicehome.com/>

Exercise 2 – Invoicing System

- Each **Customer** may generate one or more **Invoices**
- Each **Invoice** is generated by only one **Customer**.
- Each **Invoice** contains one or more **Invoice Lines**
- Each **Invoice Line** is contained in only one **Invoice**
- Each **Invoice Line** references only one **Product**
- Each **Product** may be referenced in one or more **Invoice Lines**

Note : Each relationship can be read in **BOTH** directions

Suggested Answer 2



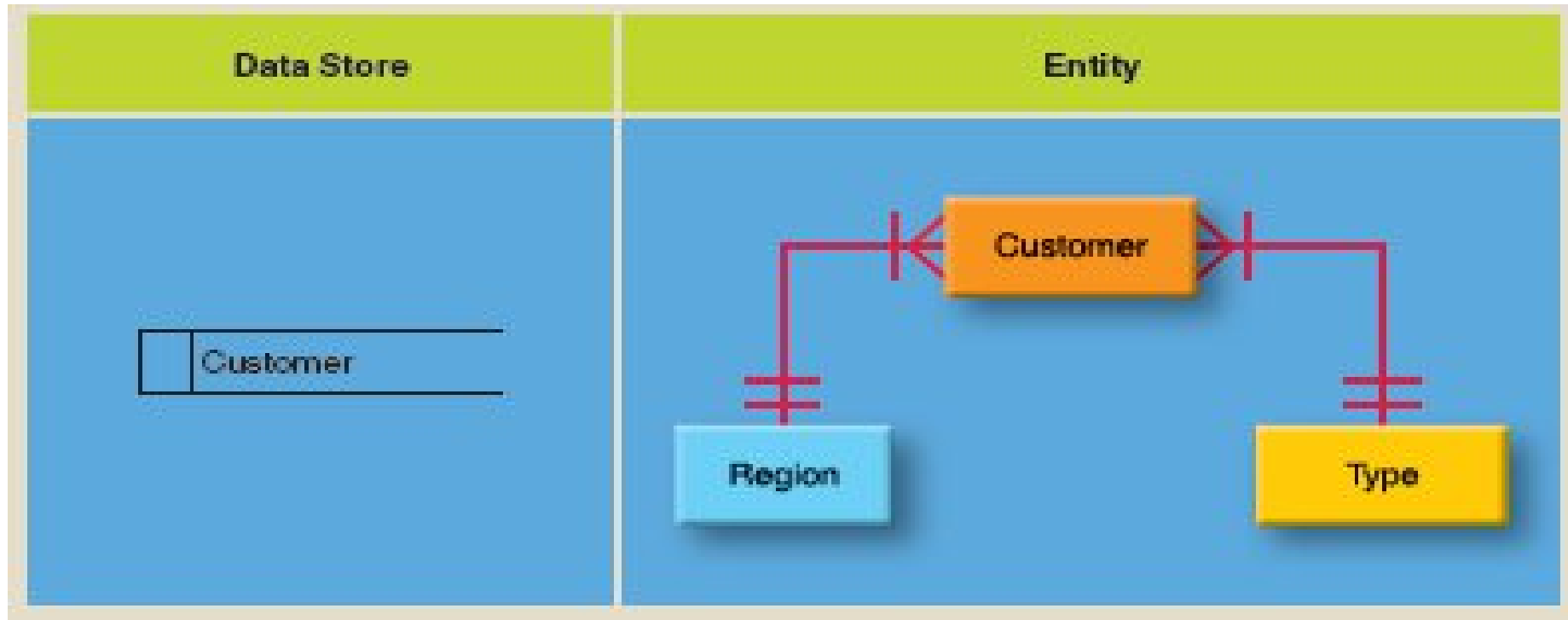
Describe the above relationships

Relationship Between DFD and ERD

- There is a relationship between DFD and the ERD. The **data stores** in DFD is related to **entities** in ERD.
- A well-drawn DFD can be used as a **starting point** for drawing the ERD.
- But take note that this relationship **may not** be 1 : 1. For example, the diary data store in DFD of a dentist system corresponds to the entities : appointment, appointment time and appointment type, in an ERD.

Relationship Between DFD and ERD

(cont...)



The **data stores** in DFD is related to **entities** in ERD. The number of data stores can be used as a starting point for drawing the ERD. But take note that this relationship may not be 1 : 1.

Purchase request (data flow --> data)

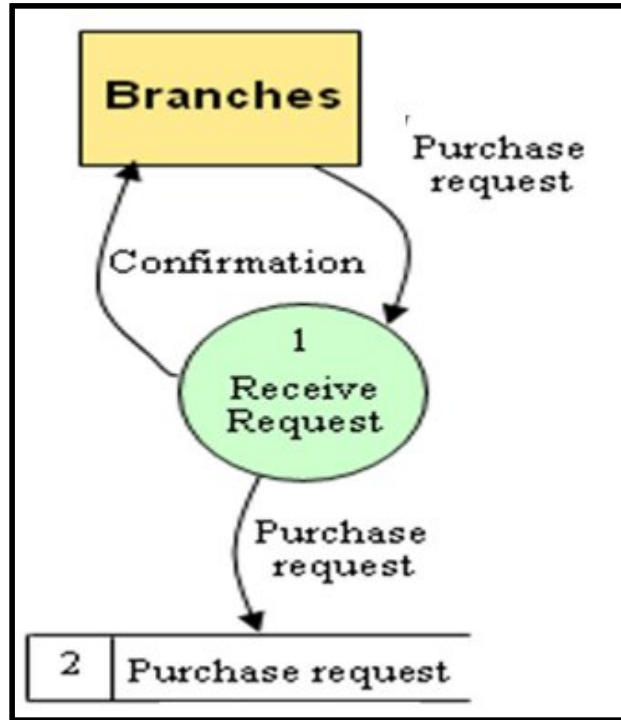
Last Monday -Perak - Ali request to buy 10 boxes - A4 papers (500 sheets per package), 1 box -4 pack	accept
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Last Thursday - Johor - David request to buy 20 boxes - A4 papers (500 sheets per package), 1 box -4 pack	accept
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Purchase Request (data store --> file)

PR01 -Last Monday -Perak Ali request to buy 10 boxes - A4 papers (500 sheets per package), 1 box -4 pack

PR02 -Last Thursday - Johor - David request to buy 20 boxes - A4 papers (500 sheets per package), 1 box -4 pack



Data Flow Diagram
(DFD)

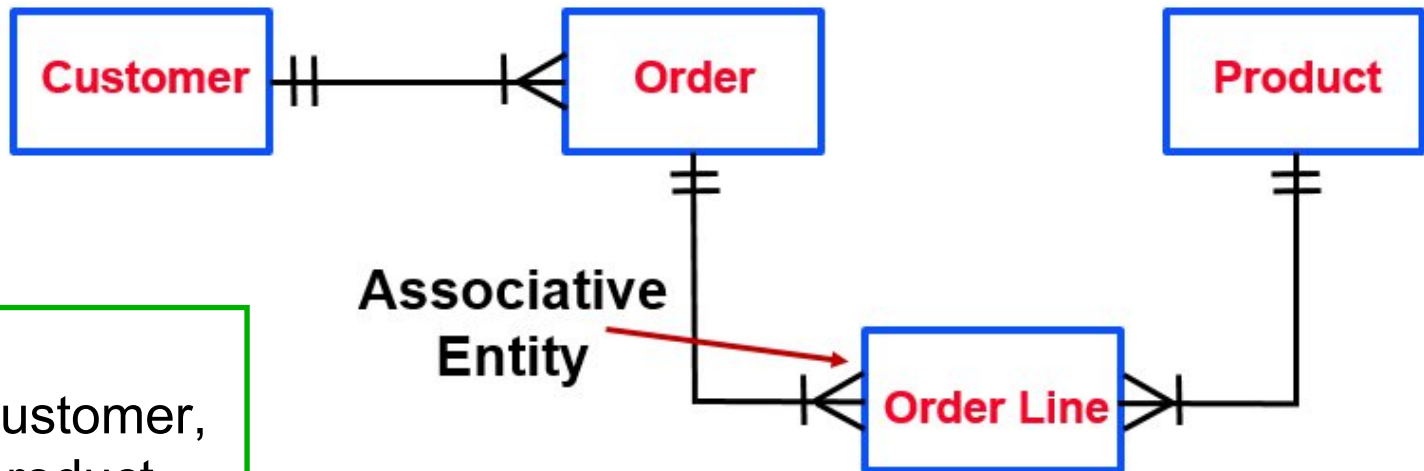
Example 1 – Associative Entity

- By introducing an 'associative' entity to form a new **2 pairs** of one-to-many relationships.

Before



After



DFD

You draw customer, order and product data stores

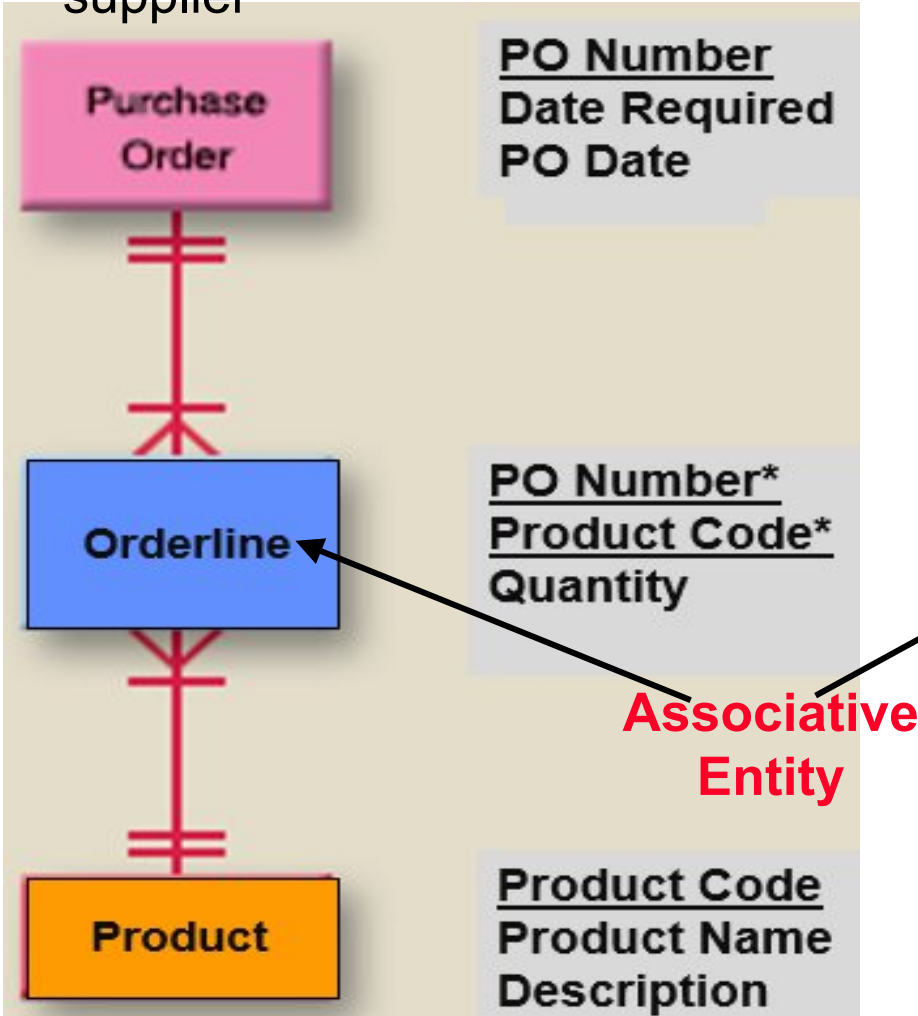
ERD

You draw customer, order, orderline and product entities

Exercise 3

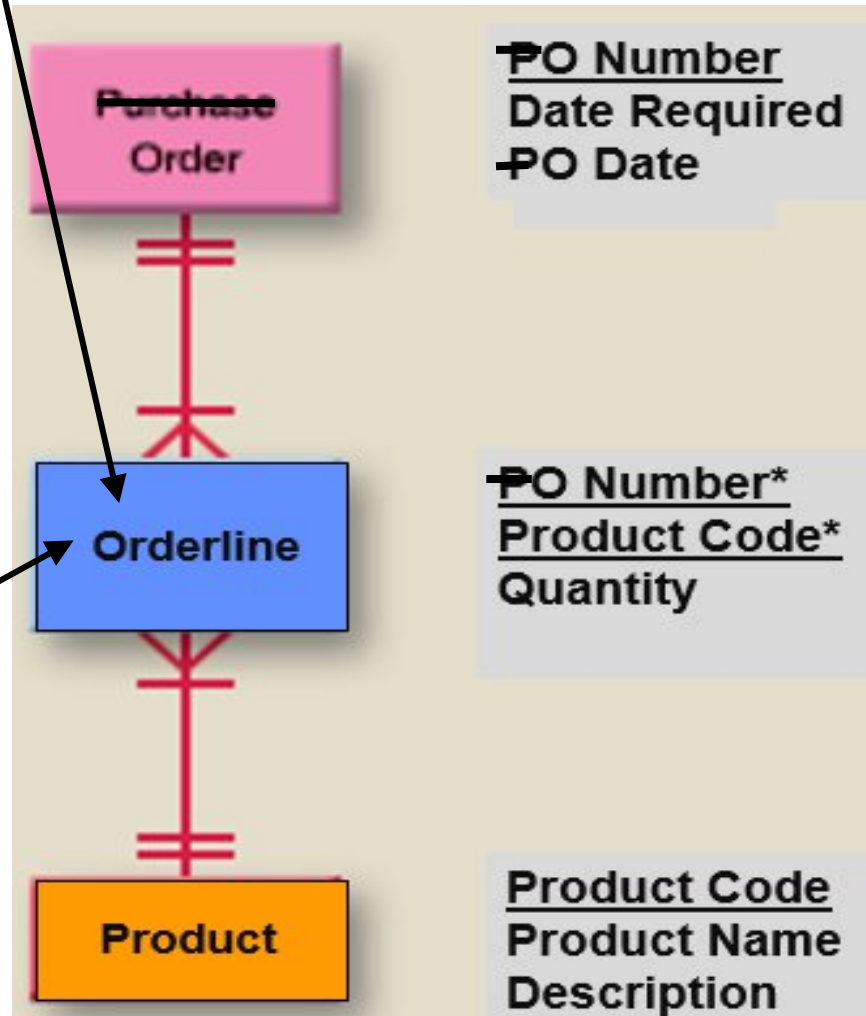
- Indicate both the **primary** and **foreign** keys

Your company buys products/items/parts from supplier



Your customer buys products/items/parts from your company

Order item



Example 1 – Sales System

Customer Buys Products /Goods/Items

KL Central Region
Selangor Central Region

order/sale/sale order
Date : 07/06
Customer :C01, Ali
Order No : **0001**
I001 item A RM10 10 units
I002 item B RM5 20 units

order/sale/sale order
Date : 08/06
Customer:C01, Ali
Order No : **0002**
I001 item A RM10 20 units
I003 item C RM5 30 units

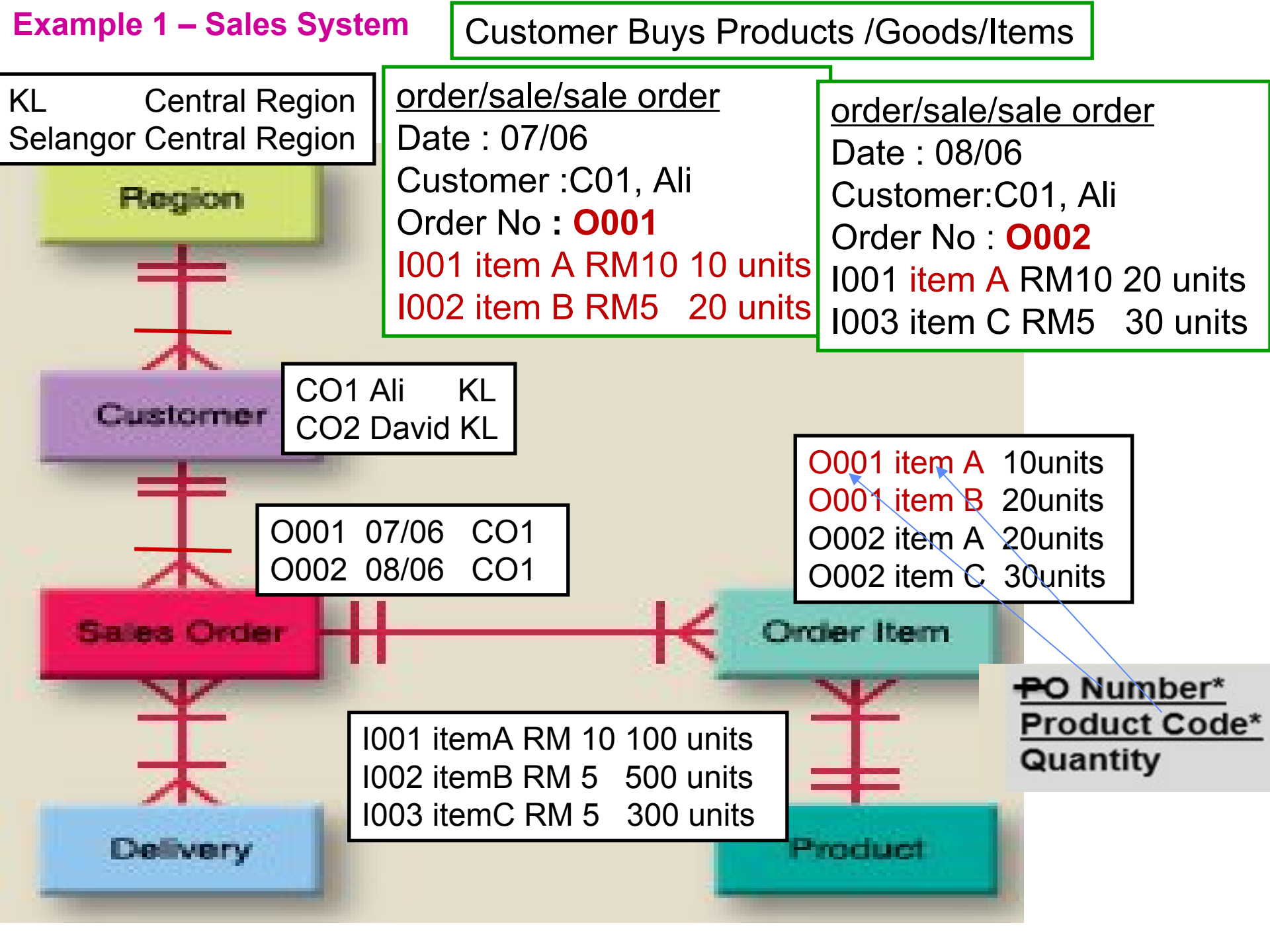
CO1 Ali KL
CO2 David KL

O001 07/06 CO1
O002 08/06 CO1

O001 item A 10units
O001 item B 20units
O002 item A 20units
O002 item C 30units

I001 itemA RM 10 100 units
I002 itemB RM 5 500 units
I003 itemC RM 5 300 units

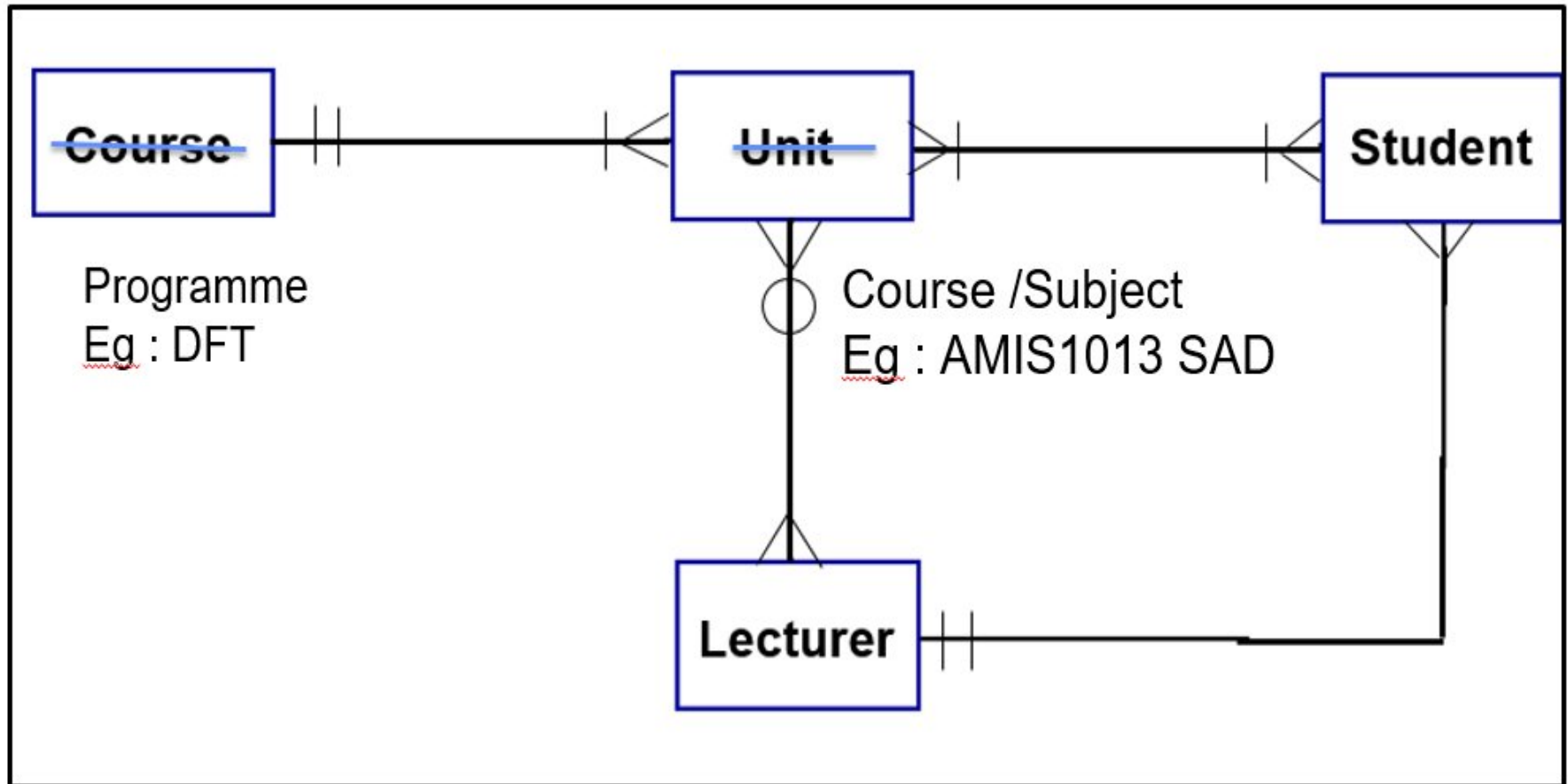
PO Number*
Product Code*
Quantity

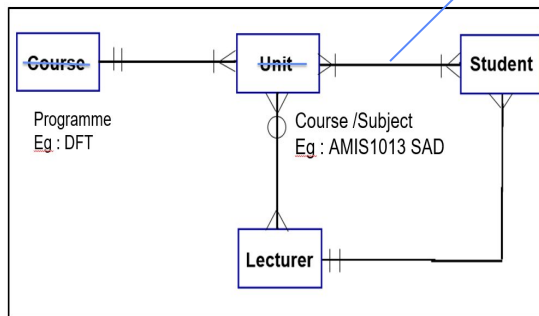


Exercise 4 for Students

- At the Best University, **students** enroll on **s** (e.g. BA in Business Administration), but they also need to enroll on **units** that relate to these courses.
- A minimum number of students need to enroll on each unit for it to run.
- Each **course** has typically a number of **units** that can make up its total number of credit points.
- Each unit is led by one **lecturer**, but can be co-taught by other lecturers. Some lecturers get research-related leaves and so they don't need to teach on or lead any **units**.
- Also, each **lecturer** may have a number of **students** who are his or her "**academic advisor**" who can consult him or her on personal and other types of problems.

Answers 4





COURSE Table

COURSE-ID	COURSE-DESCRIPTION	NUMBER-OF-CREDITS
CHM112	General Chemistry I	5
CSC151	Computer Science I	3
ENG101	English Composition	3
MKT212	Marketing Management	3

foreign keys

GRADE Table

STUDENT-NUMBER	COURSE-ID	GRADE
1035	CSC151	B
1035	MKT212	A
1035	ENG101	B
1035	CHM112	A
3397	ENG101	A
3397	MKT212	C
3397	CSC151	B
4070	CSC151	B

combination
primary key